FACULTY OF COMPUTING
UTM Johor Bahru

SCSE2123 – SOFTWARE REQUIREMENTS ENGINEERING

SEMESTER III, SESSION 2025/2026

PROBLEM SOLVING 1:

UNDERSTANDING TYPES OF REQUIREMENTS

Lecturer: Dr. Radziah Binti Mohamad

Section: 02

Group: 09

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Type of Requirements	Requirements statements	Analyst
Functional	1. The system shall allow UTM students and staff to register and log in using their official UTM email and staff/student ID for authentication.	Lee Ming Da
	2. The system shall enable users to add and manage their e-wallet balance through secure top-up methods such as online banking or UTM approved payment kiosks.	Lee Ming Da
	3. The system shall display transaction history including transaction amount, time, location, and merchant details for payments made at campus services.	Lee Ming Da
	4. The system shall support QR code or NFC scanning payment functionality at all campus facilities, securely process the transaction, and update the user's wallet balance immediately after payment confirmation.	Lee Ming Da
	5. The system shall notify users of successful transactions, low balance alerts, and failed payments through app notifications and emails.	Tan Choon Da
	6. The system shall provide administrators with access to view overall transaction summaries, user reports and system analytics for monitoring.	Tan Choon Da
Quality	1. Performance Efficiency – The system shall respond to user inputs and complete standard transactions with very low latency.	Tan Choon Da
	2. Reliability – The system shall remain available and operational all the time and automatically recover from minor failures.	Tan Choon Da
	3. Security – The system shall encrypt all sensitive user data such as passwords or payment info to ensure confidentiality.	Gan Rui En
	4. Maintainability – The system shall allow future integration with UTM's campus student card or staff card without requiring major redesign.	Gan Rui En
	5. Interaction Capability – The user interface shall be intuitive and accessible, allowing new users to learn and operate the system without external training.	Gan Rui En
Constraints	1. Legal – The system shall comply with UTM's IT policies and Malaysian Personal Data Protection Act.	Gan Rui En
	2. Technical – The system must operate only on campus Wi-Fi and university servers for security reasons.	Mohammed Alsakkaf
	3. Cultural - The system must support both English and Malay language interfaces to accommodate all UTM users.	Mohammed Alsakkaf
	4. Financial – The system development and maintenance cost must remain within the allocated university budget and utilize only approved open-source or licensed software.	Mohammed Alsakkaf

	5. Operational – The system must be fully integrated and functional before the start of the next academic semester to ensure smooth adoption across all campus facilities.	Mohammed Alsakkaf

Stakeholder	Justification
Students	Students rely on the app for daily campus activities including cashless payment at printing shop, vending machines and food stall. Students can access to campus facilities such as library and lab by using their digital ID. System also monitor students' financial status including balance, transaction history and outstanding fee through app in mobile device.
Lecturer	Lecturer use the system for attendance verification, facility access and cashless transaction within the campus. They can easily record student attendance by generating a QR code to scan or using app's integrated NFC.
IT Department	IT Department is responsible for managing and maintaining the system to ensure its consistency and security. They also provide user support, monitor system performance and resolve technical issues to maintain high availability. They are playing critical role to ensure system security and smoothness across all campus networks and devices.
UTM Vendors / Campus Merchants	Campus vendors such as cafeteria, bookstore and printing services use the Smart Wallet as a payment platform. It enables faster, cashless and more secure transactions. Their participation ensures wide adoption of the system and smooth integration with existing point-of-sale systems on campus.
UTM Digital	UTM Digital oversees the digital transformation initiatives in the university. They ensure the Smart Wallet aligns with the university's strategic goals, manage compliance with digital policies and allocate resources. Their approval and supervision are vital for deployment and system sustainability.